

# Maziar Zamanpour

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🌐 Maziar Zamanpour

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## SUMMARY

Ph.D. candidate (graduating Jan 2027) with 3+ years building optimization solutions (MIP, Benders Decomposition, MPC, dynamic programming) in Python/Gurobi/CPLEX on large-scale real-world transportation & logistics problems. Published researcher with strong algorithm design, statistical analysis, and data-driven decision-making skills. Seeking an Operations Research / Optimization internship.

## TECHNICAL SKILLS

- **Optimization:** Mixed-Integer Programming (Gurobi, CPLEX, PuLP), Benders Decomposition, Model Predictive Control, Dynamic Programming, Gradient Descent, Greedy & Heuristic Algorithms.
- **Programming:** Python (NumPy, Pandas, scikit-learn, Gurobipy), Java, R, SQL.
- **Analytics & ML:** Regression, Decision Trees, KNN, Time Series / Kalman Filter, Predictive Modeling, Data Visualization (Tableau).
- **Tools:** SAP, Git/GitHub, SUMO, VISSIM, Microsoft Office.

## EXPERIENCE

### Operations Research & Optimization — Research Scientist

University of Minnesota - Minneapolis, MN

August 2022 - present

- Built optimization pipeline (Python, Gurobi) for network design using **Benders Decomposition** on a MIP; integrated Dijkstra-based cuts to **reduce solve time by 90%** on large-scale instances (GitHub).
- Developed **real-time model-predictive control** system for vehicle red light running violation based on vehicle connectivity; deployed on public roads in Minnesota with **safety enhancement proved** (GitHub).
- Applied **gradient descent** and short-term prediction to calibrate driver behavior models; delivered individualized recommendations validated on real-world data.
- Built supervised ML pipeline to predict lane-change behavior from trajectory data, improving prediction accuracy.
- Extended large-scale **dynamic network simulation** (Java) to model energy-efficient routing and multi-agent interactions (GitHub).
- Conducted **cost-benefit analysis** for MN DOT; demonstrated payback within 3 years, informing deployment decisions.

### Optimization & Product Development — Entrepreneurial Lead

National Science Foundation I-Corps Program

May 2024 - present

- Built dispatch optimization using **integer programming (CPLEX)** and ML-based demand forecasting for car-sharing; modeled **75% cost reduction** vs. personal vehicle ownership.
- Secured **\$50K grant**; led 100+ customer discovery interviews to validate product-market fit.

### Statistics and Probability — Course Instructor

National Science Foundation I-Corps Program

May 2024 - August 2025

- Lectured in discussion sessions, designed and presented course materials problems.
- Covered topics of probability, causal inference, Bayesian method, distributions, point estimation, regression.
- Utilized **ERP software (SAP)** to initialize and configure the platform's operational system, managing data flows across supply, demand, and dispatch modules.

## EDUCATION

**University of Minnesota** - Minneapolis, MN

Dec 2023 – Jan 2027

Ph.D., Civil Engineering: Transportation (+ Industrial Engineering Minor)

GPA 3.7

*Coursework: Optimization for ML, OR for Data Science, Production & Inventory Systems, Decision Analysis*

**University of Minnesota** - Minneapolis, MN

Aug 2022 – Dec 2023

M.S., Civil Engineering: Transportation

GPA 3.6

**University of Tehran** - Tehran

Sep 2018 – May 2022

B.S., Civil Engineering

GPA 3.7

## SELECTED PROJECTS

### DoorDash Delivery Optimization

Fall 2025

- Formulated driver-delivery assignment as a **MIP** (Gurobi); designed greedy heuristic for real-time dynamic assignment.
- Developed SQL queries for data extraction, filtering, and customization to support analysis and inform the optimization model inputs.

### Airbnb Price Prediction

Fall 2023

- Built predictive pipeline: data cleaning, feature engineering, modeling (regression, trees, KNN) in R & Python; visualized in Tableau.

## PUBLICATIONS & AWARDS

- **7 publications** (3 peer-reviewed journals incl. Transportation Research Part C; 2 state DOT reports; 2 under review). Presented at INFORMS, TRB, ITS Minnesota.
- **Matthew J. Huber Award** for excellence in transportation research (2024).
- Won two **TRB Travel Award** (2023, 2025).